

Seminar on Applied Category Theory

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well let's let's get going so welcome everyone uh nice to see so many people what's online um so this is a an afternoon devoted to uh applied category Theory a lot of you in the audience would have heard me speak about category Theory before um what is applied category three is going to be the question Okay so and I'm particularly interested in getting philosophers on board trying to you know I fell in love with category Theory 30 years ago um you know when you fall in love with something you kind of want everybody everyone else should see the beauty of the thing you fall in love with uh it's been a bit of an uphill battle perhaps to get philosophers interested so when Yale University announced uh they're officially replacing the grad program logic requirement with a broader formal methods requirement that they can go off and study logic probability stats Game Theory I tweeted back teach them category theories why not you can have all of those at the same time if you just give them a basic a basic grounding accountability Theory we can do all of all of the above with just a few little add-ons foreign so yeah I mean philosophers typically through their training you know almost inevitably will encounter propositional logic uh and some first order logic depends where you are perhaps a modal logic too because they'll they'll be dealing with uh metaphysics and they want to know about the issues to do with this St possibility and so on if yeah plus some people will be exposed to second order logic perhaps some probability Theory decision Theory and so on unlikely or perhaps I mean this isn't a scientific survey I just say people will pop up and say oh yeah my department we do this this it may be but I mean I think in general it's probably fair to say that something like type Theory would be unlikely to be studied uh in a philosophical training and category Theory really very very unlikely okay now of course you can be exposed to these things in different ways and I'm kind of more interested in in bringing about quite a shift in Attitude should we say because if you approach if you if you approach things in a certain kind of way if you project yourself from the top down you know you could be exposed to category three um just by a well um yeah that's the first order logic and now I can express cash query Theory as some sort of first order Theory uh you know oh it's to do with a

collection of things a collection of objects and a collection of connections arrows between these objects and it satisfies this and this and this property so you could see it as just a sort of specific Theory uh within within a kind of existing formalism but if you do that you don't really kind of get at the heart the the heuristic heart of the matter you don't get the mindset that comes from the sort of correct instruction into category Theory and that's more of what I'm interested in so we're talking about I mean there's a slightly controversial though like this language we'll see what has come to mean this apply category Theory later but you know there have been applications of Category 3 right from the beginning very beginning so it gets going in the 40s uh within mathematics itself there's a you know there's this great effort to systematize sorts of situations when you you've got a kind of kind of entity maybe maybe like a category of spaces or collection of spaces and you want to know things about the spaces uh and they're very hard to deal with in themselves but if only you could transfer them somewhere more amenable to you maybe into some sort of sub collection of algebraic objects you can resolve questions back in the in the geometric side yeah the spatial Side by this transfer over to something uh algebraic okay so you're finding these sort of patterns these common patterns appearing between these different areas of mathematics from the 40s onwards this this becomes you know uh popular approach through the 60s I mean these are yeah these are approximate dates you'll find uh exceptions all over but you start to get people thinking you know in this category is pretty pretty good maybe I didn't need to see it uh in terms of oh it's it's it's just a kind of a categories are sort of set with some properties maybe I can make it its own foundations and there's a recently departed William levier in the 60s who who suggests exactly that the category Theory could provide its own foundations and later on in the 60s he's even so absolutely fundamental Construction in Casper the notion of adjoint Android Junctions or joint functions so this is very much when you've got the situation and you've got to sort of transfer in one direction from from one space to another space and it might not be the case that you can just reverse that process and hedge yourself back second place back to the first place but that you could do something as as good as you can it's the sort of best approximation to kind of reversing the process these things are adjunctions and Lavere discovers whether they were there even in your first order logic they were there right right and you're very very fundamental you know those quantifiers you dealt with those Universal and existential quantifiers themselves are part of that process of this this reversal okay so category through sort of seeping right into the roots of the very basic formalisms okay then computer science uh gets going over if that dates fair enough I mean maybe it's even earlier uh we've got

computer scientists here uh who can tell us perhaps about that uh and we'll kind of get a sense perhaps a little sense of why why they got interested in their physics perhaps from the 80s so we're gonna have John Baez when he wakes up he's in California so I guess you know we had to push his talk back to to five to give him time to wake up so what is it in California it's pretty early so I think he's not with us at the moment um so he wrote a paper with a student of his Mike Stay in 2009 um it's called it's a Rosetta Stone physics topology logic and computation just to see that can you perhaps the people are here a bit smaller I mean um so category theory has been used to understand uh ideas in physics topology logic and computation so it's always about you have a kind of entity of a kind and then you've got mappings between these adjectives that stuff the thing that interests you so physics you're interested in Hilbert spaces if you're doing quantum mechanics you're interested in Hilbert spaces and you're interested in operators between Hilbert spaces okay if you're interested in topology you might be uh interested in a space that has a boundary an input boundary and an output boundary and you're interested in the manifold that joins the input to the output if you're dealing with logic you're interested in having a proposition and an inference through to another proposition with computation you have data types data inputs uh various kinds people put in their data and then it gets processed and after you comes through the other end into the output okay and these parallels go on I mean so that they'll be uh you know famously with uh when you put when you put two Quantum systems together they don't just they're not just a straightforward product of their parts you can't just separate them off again entanglement ideas um but still all of a sudden you you put two Quantum systems together and you want to the the space of these two Quantum systems is this tensor product and in topology you're going to form the disjoint union you're going to put your kind of input spaces together uh next to each other and so on in logic it's exactly the same as forming conjunctions X and Y uh and in computation the product of data types and so on and so forth so you're getting these these very similar ideas of just being manifested right across the board okay so that we've got this uh fun expression at a certain point um they've got a bit controversial somebody wanted to call it computational trinitarianism and some people didn't like the kind of spooky theological tone to it so it sometimes gets re designated as computational trilogy uh okay but this idea that these things are just part and parcel so it's not just that there's an analogy going on it's that they really are three faces of the same thing called the father son who only goes to kind of idea okay so computation spaces and logic they're just they're just three size of the same thing with the thought that therefore if you make an advance in any one of these areas they don't to manifest

itself in one of the other other areas it's really that tight if you see it from this point of view okay now that we're going to still have to head off and start a different direction very good things are still going on devoted quite a lot of my life to this sort of thing so book 2020 model Copy Type Theory suggesting that um a very good thing if if philosophers got on board with this language yeah all sorts of ways I think philosophers could use this this these kind of ideas I mean if people know there's people work on category mistakes uh Gilbert Wiles ideas have been updated I mean everything the computer science scientist wants to do by controlling input data to be of the right type and this is so close to the kind of idea of category steps that Ryan was worrying about back in the 30s and 40s uh yeah I'm gonna give you another talk in a couple of weeks on on I mean it really lends itself the the kind of Brandon Robert Brandon's approach to to inferentialism I think it could be beautifully realized in this program but that's not for today um yeah another recent bit of work I did which was which was really trying to sort of Chase up the the passage the path that takes us right up to you know the kind of cutting edge of category Theory uh and type Theory means physics it's something expressed in this paper here which is is kind of pointing to this work just to point out so this this kind of work goes on that's really really kind of cutting-edge model type Theory meets physics right up in that highest highest Heights of theories of quantum gravity and whether you think that's going to work or not but it's coming much much closer down to earth now some of the work they're doing out there which is is looking at Quantum compute computations down to earth but it's a closer to Earth in quantum gravity anyway uh using using this this linear homotopy type Theory the linear of linear logic but again this is not today's work because we're going somewhere slightly different so yeah the further set of applications starts to take place let's date it from the 2010s although maybe you can again you can plug precursors and what's that looking at okay well these are people scattered through the country uh I'm sorry through the world but pretty well represented in the UK [Music] but they're looking at things like causality probabilistic reasoning statistics learning theory deep neural networks dynamical systems information Theory database Theory natural language processing cognition Consciousness systems biology genomics epidemiology chemical reaction networks Neuroscience complex networks Game Theory robotics Quantum Computing and the list goes on okay so we're away now from the very sort of high-end physics applications and we're dealing with some you know some some sort of things that of course philosophers should be very interested okay and and in a sense it shouldn't be such a surprise because I mean you're dealing with but if you notice back there that list they're dealing both with physical processes the various natural

processes out there in the world and then they're also dealing with inferential processes okay so kind of interestingly uh the category theory is good at dealing with both because they're both to do with movement of certain kinds of starting place to a finishing place if you want to go all hegelian on us you're kind of unifying that subjective and objective logic people tend not to use hegel's language these days and yeah I mean something all over the place uh these these string diagrams if you oops just click anywhere any any sort of Applied category Theory you just get this end of this beautiful diagrams these 2D grams lots of wiring diagrams going on reverse clients wiring things with nodes um right across the board and in fact this is this is kind of one of the ways that uh well there's any I think John buyers will tell us about it at five o'clock but he he was just fascinated by the way you found these these very similar diagrams across engineering and chemistry and biology and so on and so forth and there had to be something in common that unified it and he thought everything would be the common Essence to why these diagrams are all appearing um across these different domains okay so you're going to find these all over the shop okay A bit of people are very interested in this compositionality that that links you off there's a journal now called compositionality which is a general devotee to this applied category Theory and why it's so appropriate is well there's various things you want to do some very basic moves is is um I mean we actually already saw it in that Rosetta Stone thing the idea that if you if you have one system at some process uh of some kind going on that takes you from a beginning to an end we can put together two systems in parallel okay we can compose systems in parallel and we have to you know again diagrammatically it's all very obvious why you should do that you're just you're just sort of joining two two diagrams and you put it them next to each other and you don't need very much compatibility between the systems to do this if on the other hand you want to put them in series if you want to make one process feed into the second process you're gonna have to make sure that the output of the first one matches the input of the second one okay or they're not going to join together again very very category theoretic it wants it wants its inputs to match its outputs yeah or there could be something more subtle going on there could be sort of several inputs and several outputs and maybe only some of the outputs being matched by the new input to some other new device and you could just plug in those wires that match into the new system yeah or sometimes you can take a wire coming out bend it round that could be a useful move you might want to make as well uh sometimes also you've got a system with a networking system and you've got a little chunk of it and and you want to replace you want to take a little box inside uh that's that system and plug in something more complicated with the

right inputs and outputs so you can kind of compose by plugging one system inside another okay so um this is I'm sure we're going to be hearing a lot a lot about this over the next two hours and a bit so Toby Toby Smythe's uh talking first I'll give John Bice bit more of an introduction when it's that time for him um so I was intrigued to you know a Toby's work um looking around some of the papers he's made available and this this is his default thesis so it's called just last year I guess is there 22. mathematical foundations for a compositional account of the Bayesian brain and we'll be hearing about that today so primary motivation and writing this thesis is to lay the groundwork for well-typed notice cognitive science and computational Neuroscience so this is this is pretty high pretty intricate interesting stuff this isn't just you know simple physical processes just going around in their simple way it's it's tapping into a field that I've been looking into a bit of late this um this world of uh predictive processing active inference you know the free energy if we had sort of Carl Frist in his big name in the field um this yeah some interesting stuff going on there kind of hierarchical Bayesian models involved in the way that we deal with the world we make all these predictions about the way that we you know most of the world is is sort of fantasized by us it's just a slightly constrained by our influence in some sense so there's all sorts of ways this very strong top-down component to our perceptions uh interesting stuff so curiously I find myself um it would almost plug the Gap so I've you know I've spent a lot of my life as I say if they're interested the category theories type Theory could spell being interested in machine learning bayesianism I had a formal life that kind of pops up from time time occasionally uh which was an early interest in psychoanalysis and then I wrote a co-wrote a book called psychosomatic medicine so sort of extraordinarily you could it could even be this kind of bridge that takes place between those um so there are plenty of people that are there's a lot of psychotherapists now talking to this active inference approach to the mind you know the brains are kind of optimizers the trouble is that very often they get kind of caught in sub-optimal optimizations you can get locked in in a certain kind of way and the heavy top-down processing sort of makes it very difficult to sort of shift shift the system and gets kind of locked in a certain certain way I mean you know for example delusions for example uh cases where there's such a sort of strong top downness to the to the sexual apparators that the input from the world does nothing to constrain or not enough to constrain from the delusions so it's a very extreme case but all sorts of psychiatrics light um yeah so gosh if if Toby can help us sort of plug the Gap here I think we can be kind of fun if I could if I could bridge the gap of my interests so I'm certainly very much looking forward to hear what he's going to say today I noticed this rather

longer bit it's taken from his thesis this well-timed cognitive science and computational neurosense and and it's got yeah so types types of what render these these Concepts precisely allow categorical models to be cleanly compositional so it's again there's very much things I've been talking about this sort of plugging together if the interface is right so I think this is something that maybe a from what I've seen of the active inference literature well I've been intrigued to see if if he can make that something more category theoretic meet the active inference I think this is a really exciting project okay that's uh yeah I think that that brings us the end of the introduction because uh we're keeping rigidly on time and here we go it's 10 10 to 10 to four so let me close down my thing uh stop sharing that's again can you share Now Toby how's that uh yeah that's that's good do we get that full screen here what do we um this is so useful uh we can get it mostly full screen I think I'm just getting the we can do that and we can do that and there's one more thing I can do what's the other thing well there's probably enough is it that's probably enough I don't think I can do it in the browser one okay yeah that's brilliant great okay so my great pleasure to to introduce Toby Smythe to talk to us on Bayesian brains cyber categorically great thanks David and thanks to the very warm introduction um I'll try to live up to it um so my my yeah so my first degree was actually in philosophy and psychology and I guess it's possibly thinking also my enjoyment or predisposition to Vicky about things like the mind in this kind of abstract way that possibly philosophers are kind of um you know you know inclined to is What's led me to enjoy using this language that David um so clearly loves using as well um I'm not going to be saying very much on the sort of philosophical side today though you know I yeah I believe as David does that this language has a lot to to give to philosophy what I'm going to be doing really is um to talk a little bit about this journey that David nicely summarized the beginning that I've been on um basically trying to replace one kind of confusion about what's really going on with this idea that's called predictive coding in the computational Neuroscience literature and the sort of related set of ideas called active inference um with a different kind of confusion which is all about you know category Theory and this sort of higher dimensional algebra that goes along with it um I I suspect I'm sort of I'm in a happier place with that second kind of confusion it suits me better but I also think that it it helps us to understand a kind of idea which at least as far as I was concerned wasn't very well wasn't wasn't crisply or clearly expressed so far in the literature that had been written in computational Neuroscience possibly because yeah they're the interest is in specific models or specific bits of the brain whereas my interest is really in just you know what brains do or more generally what adaptive systems do um and what's emerging now in

the apply category Theory World bringing together all those different domains that were listed on that slide earlier um is this kind of this idea this this thing called categorical cybernetics um it sort of uses this kind of nice General language of category Theory to bring together ideas in in you know economic theory and machine learning and reinforcement learning and computational Neuroscience and show what are the commonalities and allow people as a result to talk to each other a bit better and to understand what they're really thinking about in terms in each other's language a bit better and so what I'm going to do is maybe sketch some of this I'm going to try not to assume too much you know background in either Neuroscience or in mathematics or particularly in category Theory um I'm going to talk about this idea of the Bayesian brain and this related idea of predictive coding I'm going to talk about um kind of current understanding of probability Theory using this compositional language I'm going to talk about Dynamics and I'm going to talk about how we could see predictive coding is basically a way of translating certain kinds of statistical models expressed in this first compositional language to kind of uh compositional dynamical systems that are expressed in the second language and so when David was talking about how categories are really useful for talking about things like translation between you know different mathematical languages that's really what we're making use of here okay the first thing I want to do though is to say a little bit more about what I mean or what people mean when they talk about the base of your brain um and this later set of ideas called the free energy principle and predictive coding um it's a very sort of prominent idea in computational Neuroscience um and you know it may be a slightly controversial too but it's it's a claim that most of what the brain does um and indeed sort of most of what's kind of living systems or adaptive systems do generally can be understood as approximate Bayesian inference and so what that means is that you have a system and it's it's trying to predict what's going on in the world more precisely maybe protect predict you know what it's it's what What sentences receiving and then on the basis of the sense data it does receive update its beliefs about the structure of the world or what's out there in order to make its predictions um better or more veridical um and this thing called the free energy is a quantity that you can optimize um in order to do inference of this kind you know kind of better way or to improve your doing of inference and if you if you minimize this quantity at least in certain contexts you get dynamical systems um which look a bit like dynamical systems that people had already written down earlier in the literature in computational Neuroscience that relate to various bits of say the visual cortex and and so that alignment says you know it says to a lot of people okay well maybe this is what those circuits

are actually doing um one of the difficulties you get in if you're just going straight from you know the statistical model that you might be interested in to dynamical systems you're going very manually um is that there are kind of a lot of moving parts and particularly if you go from one situation to another you might have to Red arrive all of your Dynamic equations again um even if when you look at the brain it has some kind of nice modular structure and if you look at these statistical models so they do too and this is a kind of hint of compositionality and so it seems to be kind of a kind of neater thing to do to see what the essence is of this process and to try to you know make this translation once in a compositional way so that you don't need to do it over and over again and you know end up with very sort of impenetrable papers full of difficult equations you just end up with in financial papers that require you to know a language that you maybe didn't already know but I think what I hope is that people start to learn mathematics in a in a compositional way in the future and so that these these papers or this work that is expressed in this category theoretic language um doesn't you know scare people off in the way I'm gonna sit here this might do now I can hear some yeah online is that you John could you please yes that's right thanks um okay so I mean I don't want to spend too long talking about why category Theory because David's done a very good job already um I'm just gonna say some sort of you know some general say some generalities and hopefully they sound nice and convince you that this is not a crazy thing to be doing that category theory is like the mathematics of composition of sort of pattern because it's about translation it's about interconnection it's kind of about metaphor too it's about saying okay well here's this pattern it crops up in lots of different places and maybe this is like this other thing and that's kind of like a kind of translation and as we've been indicating that it seems to promise to supply a kind of lingua Franca for mathematical science generally so that you know each different sort of category of systems could be like a little library in this in the sense of computers and computer programming that you can import it and have access to all the kind of nice tools that are available but because it's in this common language they all these different parts plug together very nicely and that's you know this happens because categories they collect into this bigger category of categories so they it's that kind of supplies its own foundations as David was indicating and this allows us to use the morphisms between the categories and themselves these are known as functors to translate from one of these libraries or mathematical universes to another um thereby sort of connecting them together um but so that's kind of like one major benefit of using this language and another one is that it kind of because you have to say like what are the interfaces of your system like where do they start and where do they end or you

know what do they connect from and to it kind of enforces this discipline it kind of enforces a kind of clear thinking it enforces um an effort to carve nature at its joints in this kind of platonic way okay so to make a little bit more progress at least with the mathematics I want to try to say briefly what a category actually is um it's a collection I often denoted with $\mathcal{C}$ for category like this kind of with this little tick in it it's a collection of objects so these are the things that are being related often denoted c0 and then for each pair of objects you have a set of ways of relating them to each other it's a kind of it's a set of edges or morphisms um so if you write them like this it goes from A to B and the key detail is that if you have away from going away from where to go from a A to B and then B to C as I've drawn down here you get a composite morphism like this and so it's a bit more than I mean it's quite a lot more than just a graph it's it's a graph where the edge is compose and this is where the compositionality comes from um there's also an important extra detail which is that each object is witnessed by amorphism of its own called its identity and then you've got some axioms that say okay well this composition is associative and it's unital with respect to that identity um those kind of details aren't super important for what I'm saying today but I think I just want you to have in your mind this notion of compositionality and particularly uh compositionality of morphisms once you've got this structure you can start to study it in itself as David was also talking about and one of the things you can do one of is prove this fundamental theorem and it has a sort of quite in some sense trivial but very deep and sometimes confusing proof um it's called the United Lemma and it and it says something that's very familiar to us in everyday life um and this is a kind of um a maxim from a linguist I think called frith um who said You shall know uh a word by the company it keeps and the United Lemma generalizes this to all the these categorical contexts that you shall know an object by the company it keeps in the sense that if you know how a thing is related to the other things in its new year then you can identify the thing itself and that you know we know not just from Linguistics but if you know how to get to a place from the places around it then you can you know figure out what where that place is on the map so you can sort of identify that place um and it's a very it's a kind of very it's a very powerful result and it's a kind of nice kind of nice evidence that category Theory captures something about the way the world is built up but the thing that you know I want to talk about really today is the brain and uh using the tools of categorical cybernetics to model it um so the brain is this nice complex system it's made up of other complex systems so it's made up of neurons which of course have a lot of structure of their own and it has this kind of computational behavior it does prediction and it does planning and it does all these things that

we have programmed computers to do now um and you know brains themselves compose into bigger systems societies so it has a lot of features that make it you know that seem to suggest it should be amenable to description to description in this categorical language um and indeed it should be comparable to other other systems that people have started to study in category Theory or in applied category Theory cybernetic systems um this really I guess at least in this form kicked off with the the um introduction of compositional Game Theory by Jules Hedges and his colleagues um and so there they study economic um and those are very closely related to systems of reinforcement learning um and indeed structurally also related to systems in machine learning as it is in the form that you know is given rise to things like chap GPT and these things all have a similar kind of structure there's some kind of process which might have some kind of control input and then you know there's another part which goes backwards a kind of retroductory part a sort of feedback and the process and possibly the control and the feedback you could think of them as all being contained within one system that is in kind of traditional cybernetic maybe called a plant or in control control theory called a plant but this is really just like a part of a bigger system it's open to some environment and the brain has you know similar structure at least in these kind of visual cortical areas that have these predictive coding model models that I um introduced at the beginning and so there like a little bit of neural circuitry maybe has one part which points from deep inside the brain here to out towards the world here and it does say prediction about what what you know the eye should expect to receive and then there's another set of neurons which goes from nearer the world to deeper inside the brain that that feeds back the error in that prediction and allows the brain to form a sort of updated or new belief in Bayesian um probability it's called the posterior belief and so you have this kind of bi-directional process which looks a bit like a kind of cybernetic Loop and so I'm interested in not just the brain of course the brain is a great Target and obviously I I um you know part of me as a brain I want to understand that um but I'm interested in systems that do this kind of thing adaptive systems in a sort of General way and because of that um I'm I'm Keen to use these General tools uh and to study this system that is so good at doing this kind of adaptation which is the brain okay and so you've seen me draw on the previous slide a diagram and it's a diagram of the kind that if you know about category Theory or you did some searching as David was suggesting is known as a string diagram and so I want to introduce categories again or at least composition again using this language so that when I use these diagrams again a bit later you'll know what I know what I'm doing the great thing about using string diagrams is of course for most of us we have this very well

developed visual and so we can not only use category Theory to reason about it but we can also use it to help us reason about mathematical things like categories themselves in particular you know when we use pen and paper it's a two-dimensional medium and it's a kind of often for processes where things can go in parallel as well as sequential sequentially it's often more convenient to draw the processes out as we see when people draw other kinds of more informal diagrams than it is to write them in this kind of more sort of traditional algebraic symbolic way that you could write on on that I've drawn on the right hand side and so what we can do is if we have a process F that goes from A to B so we're going to start by reading on the left and then if you have another process G that goes from B to C we can plug them in and so then you have a process that's G after F which in traditional mathematical notation is denoted like this and so that's like the sequential composition and we can also put another process H from X to Y by the side of it and so this thing now goes from the tensor a tensor X to C tensor why um and you have as I said each object has an identity morphism and this is just what the string itself represents and you can put this next to you know processes and you know you can use this to plug things together um you have special processes that start in like the empty diagram which we write with this I and they we think of as like States so in in physics they would be like vectors or in probability Theory they're like distributions um and then you have things that go the other way that sort of terminated an empty diagram and these might represent like say uh predicates on some space or they might represent effects that you could measure or measurements on your physical and these are kind of morphisms in the other direction in into this other into this unit diagram into this unit object and with these diagrams UK you can not only sort of depict the processes themselves but you can write down equations and start to manipulate the diagrams accordingly and so here at the bottom as an example I've written the equation that represents the co-associativity of um of copying so it says if you copy once and then you copy one of the copies it doesn't matter which one of the copies you you've copied and you know this has this nice kind of Fairly obvious graphical representation here which is a bit more I think natural to use at least for this kind of thing um than the sort of algebraic sort of symbolic notation that maybe is more familiar to people from mathematics one of the things um that we can represent in this language is probability Theory um and this you know the kind of natural kind of length you know mathematical formalism to to use for that is that of Markov kernels which you could really think of as like conditional probability distributions and so here I've got this thing C it goes from X to Y and we could think of this as like a pro conditional probability distribution of Y given X but now we have them these things

in this compositional language we can plug them together so if we have one and then another we can compose them together by sort of marginalizing over the middle thing and this says okay well we'll sample an output we'll sample an output from Y and then we'll give it to Z give it to D and then we'll run whatever presses D represents and that'll give us something in a distribution in Z and so this is like some something it's like d after c Z given X we can do copying in probability two and this is like you know it descends a thing to the Delta Distribution on the copy x equals x Prime which you might be familiar with and so here we can and we can use this to formalize in this kind of graphical language joint distributions and so here we've got a joint distribution with one thing on with one module on X and a kind of what's called a probability a likelihood which goes from X to y and this thing depicts the kind of this sort of um this representation of the joint distribution is like a likelihood a conditional probability distribution on Y and a prior and this is the kind of one fundamental part of Bayes law that you might be familiar with and as I indicated on the previous slide these kind of co-states or effects they represent things which we could think of in probability as like fuzzy predicates um and that might be something that if you're philosophically inclined you've thought about we're not going to use them very much but we will use them to represent loss functions in machine learning and in predictive coding later on so at the heart of this kind of uh predictive coding idea is Bayesian inference and what Bayesian inference does is effectively takes one process here this goes we're reading the diagram upwards now just you know first reasons of space here this goes from X to Y and so we can think of this as predicting something in y given something in X but you know it's a kind of stochastic prediction it's fuzzy and so Bayesian inversion allows us to go to go instead of from this process which goes from X to Y to a process which goes um from y to X here and so this is like reversing this process it allows us to say okay given what I observed what should my belief what might be a better belief have been and the the Criterion that it satisfies is that the kind of these are two ways of writing the same joint distribution so you start with a prior belief that says okay well this is how I think the world is and then you have this way of generating data and this says okay well this The Joint distribution you get over the say causal structure of the world and the observations is the same whether you go forwards or backwards and so this this kind of makes it like a well-behaved inversion so it's a kind of it's a kind of fuzzy way of inverting this this predictive process and Bayes law is often written like this thing it's just a way of expressing the conditional probability in the other direction and it just comes out of the kind of laws of probability that you might be familiar with um this is the product rule of of

distributions but I'm not I'm not here to give a tutorial about um you know about fundamental probability Theory I'm going to try to build up to predictive coding I've start I've already introduced categorical probability Theory and I want to get from there in the kind of piecemeal way from like simple steps using like simple you know simple like compositional Parts towards the dynamical systems that represent how the brain does this inferential process actually dynamically um and so as I indicated you know a brain it has lots of parts that may sort of plug into one another in particular it has some parts of the brain which are nearer the retinal the eye and other parts of the brain which are kind of what we think of as more sort of high level that represent you know objects or concepts rather than say like what we might think of on a computer as pixels and so some somewhere in the brain you have to go from these kind of low level features to the high level high level ideas and this happens in this kind of hierarchical way so zooming out piece by piece and so you know you might imagine being say at the cinema and you're sitting right at the front and you're looking at you know looking at the screen but you can only fit in your visual field say a part of the screen at one time but yet you can sort of keep track of what's going on everywhere and so you maybe have this process where you've got you know a belief about what's happening on the whole screen and also another process where the eye is receiving signals from part of it and if the first thing that happens is your brain updates the belief about what's going on in that little part and then it can update using that new belief yeah the you know the sort of higher level belief about what's going on on the whole in the whole scene or in the story more generally and this is a hierarchical process and really it's got this kind of compositional structure and so we might wonder okay given in this kind of abstract language a sort of Kernel from X to Y and another one D from y to Z what its inversion is so what is the inversion of the composite Channel or kernel um and how does it relate to the inversions of the parts and so this is a kind of question about the compositionality of Bayesian inversion and it's not only relevant in predictive coding of course it's relevant in all sorts of contexts particularly in um you know say things like uh probabilistic machine learning and things like that and so it's actually quite easy to show that the composition of of um the inversions gives you the composition of but gives you the inversion of the composite Channel um an important thing to notice from these pictures is that the inversion say if we just go back to this this this kind of defining law this is what we could think of as a kind of abstract base law that the inverse channel here the whole law in fact but particularly the inverse Channel depends on the choice of Prior because if I took if I change this if I change it to Π prime it wouldn't necessarily satisfy this Law And so there's a kind of

dependence here you see it comes in in the in the in the sort of defining equation here you've got P_i popping up and so we can't just sort of in some sense reverse the channel directly we need to say okay well what is this prior that I'm reversing it with respect to and so we'll see okay well if we use this law twice we can you know invert first of all d but you see we've got C coming in first and so that the the prior for this inversion of D is C after P_i and if we invert you know if we invert if we invert oh sorry we should invert C first the prior is P_i and then we invert D afterwards but the prior is C after P_i we get this composited version over here and so this says okay well we can decompose the Bayesian inversion of a composite channel in terms of the inversions of the factors here and here and in particular this kind of dip this kind of structure this dependence what is this this thing how does this arise if we try to package this up neatly we will end up using what is called a lens and this is something that's already been used in categorical cybernetics and in categorical dynamical systems theory um and then all sorts of bits of category Theory already to formalize these kinds of binary processes and so it's very nice and so maybe somehow mathematically inevitable but this kind of formalization also happens to be the way that Bayesian inversions compose and so we can say a little bit about so I've indicated that these inversions see this is the inversion of the channel C on the previous slide $c \times$ to y the inversions they depend on the priors and the prior is a state on the domain of the channel and so we call the inversions which we might write to see it's dagger for you know technical reasons it's it because it effectively makes up what's known as in other bits of category Theory a dagger functor but this thing is like a state dependent Channel and so it takes in a prior and it gives us the channel in the backwards Direction so this is a function from states to channels like this and in fact we can compose these things if we have a state dependent channel from A to B and it depends on X and another one from B to C that also depends on X we can copy the state that we depend on and give it to both parts and then compose those parts together and that you know is something that should be hopefully quite natural to see in this diagram if we sort of follow along the flow of the information through it and this means that the state dependent channels on a particular state so these all depend on distributions on X but these things form a category and we can you know write down the morphisms the set of morphisms in this way so not only though do the state dependent channels on X form a category for each X but also if we have a process you know from y to X we can say re-index these these State dependent processes so now we can pre-compose the input to F and F Prime and G Prime over here by say this morphism from y to X so now if we have a state on y we can also feed that through p and then feed the that composite State into F

Prime and into G Prime and still do the same composition and so this gives us a mapping indeed a functor from the category $\text{stat } X$ to $\text{stat } y$ and so that's going to be important because as we've seen here it Maps like but what we did before is we had an inversion on D and we pre-composed it with C to give this thing $D \text{ dagger } C$ after something and so this cap this with re-indexing captures what's going on in the composite inversion story and we can sort of you know use category Theory or categorical tools to package all of that stuff up very nicely we can pair an individual predictive C with its state dependent inversion $C \text{ dagger}$ and the pairing of a Channel or kernel with an inversion which may or may not be the exact Bayesian inversion that satisfies that equation but a pairing of that kind is what I call a Bayesian lens and I don't have time to explain how we obtain the category of these things but it is obtained using a sort of canonical tool of category Theory called the growth and deconstruction so we have this index category $\text{stat of State dependent channels}$ a family of categories for each possible Space X that's compatible with this re-indexing if you have an index category of that kind you can apply the Groton deconstruction and that gives you another category that is the the morphisms which are pairs like this and there's a kind of composition rule that comes out of the grid and deconstruction which tells you how to compose these lenses and it in fact captures exactly what's going on with Bayesian inversion and so Bayesian inversion at least it's compositionality has this nice in some sense Universal Property but because it because it arises in this kind of abstract way it's not an ad hoc thing that depends on the the situation at hand or the kind of problem you're studying it's it's a kind of mathematical consequence of the structure of of you know what's going on for in Bayesian inference and you know we can use string diagrams to really spell out this compositional structure and it captures the picture I drew earlier of this predictive and updating process so now we have these two two little lenses here and here and so one now you know goes from X to Y and we can think of this as doing like a high level prediction and then here we've got another goes from what y to Z and we can think of this as being this lower level prediction and we can compose those things together to get a prediction that goes from the high level to the lowest level and we can also follow the information flow backwards so now suppose we've got some feedback from the world here well okay we flow through the forwards information and this gives us the kind of prior the pro C after π and then this gives us an updated intermediate level belief and now say we had Π here now the inversion here takes in π C Prime π and this gives us the higher level belief and this this thing here is exactly what we had from Bayes law and so this is what I mean when I say that you know the Bayesian updates compose optically they compose according to this

lens pattern and as I've already indicated this this comes up a lot in categorical cybernetics in fact this kind of chain rule is exactly how that propagation works and it's it's how sort of utility is passed back in economic games um and so it's interesting that this this kind of nice General construction captures all of those of all of those um all of those situations um and in such a way that actually applies to this physical system that that is the brain so I've said that there are some of these lenses you know $C \dashv C$ dagger where this is some State dependent channel that actually satisfies Bay's law as I drew it a couple of slides ago but there are lots of other lenses where C Prime has this type the same kind of type say it goes it takes in oh sorry takes in a y and it depends on some prior in X and it gives you some new updated X there are lots of lots of State dependent channels of this form and they're not all exact Bayesian inversions we could think of them as like approximate inversions the way that we could formalize this chain rule in in the kind of neatest way of exact Bayesian inference is as what's called a section of this vibration so Bayesian lenses form a category where the morphisms of these pairs and we can sort of project out we can forget this part just to end up in C and a section is effectively a mapping from C into this vibration above it such that if you do this round trip you get back where you started and what makes a mapping a functor is that it preserves composition and so what's going on here is that we map C to $C \dashv C$ and we also map the to $D \dashv C$ such that if we map DC we get the the inversion of DC which as we've already seen is $see \dashv C$ after $d \dashv C$ and so the fact that this this mapping of channels to lenses where the second part is the exact inversion is a section exactly says that Bayesian updates composed according to this law now again this isn't the only section of Bayesian lenses there are others and those ones represent approximate inversions that are particularly well behaved because they satisfy this compositionality property and I call those things inference systems and finding an inference system is like the first step in in sort of defining the approximate inference process that is predictive coding but you know once you've got an arbitrary inference system you you know it might not be very good at actually doing prediction or updating of beliefs if we think of Bayesian inferences like the optimal such process and yet nonetheless you know Evolution hasn't just picked brains to be instantiating some like arbitrary inference system it's chosen ones that are actually quite good at doing prediction and updating and so there must be something going on that allows Evolution effectively to measure how good a system is and that's what natural selection is doing um and similarly there must be something on something going on that allows brains to improve their predictions or effectively to learn um and what I'm going to talk about now is how we can equip each Bayesian lens with a loss function

effectively a way of measuring this this goodness of a prediction um but because we have this compositionality hanging around it satisfies some nice you know some nice constraints that is enforced by biology which is that neural computation is what you know what biologists or neuroscientists say is local um that means that the the data required of the computation must be available to the neurons themselves it shouldn't depend on something that's going on further away in the brain the information has to be accessible in order for the neurons to actually do the computation they need to do and so because we have this compositionality if we enforce that of say the loss functions or of the lenses as well that it that makes all of the the information local to the individual parts of the system and so that makes it a kind of nicely biologically plausible thing which is a bit different from what happens in say back propagation machine learning where changes to some parts of the system may depend or like the way you measure how good the performance is sort of filters back to the whole system at once the loss functions there aren't necessarily compositional um another thing that's important to bear in mind at least um when it comes to approximate inference systems is that their performance kind of depends on the problem they're trying to solve so obviously Evolution has selected different organisms for different niches and so you know if you change an animal from it one Niche and you put it in a different environment it might not survive as long and that's because you know that it's you know we could say that abstractly that's because it's inferential performance is kind of context sensitive it depends on the beliefs it has about how its environment is structured um and it also depends on like how it actually you know absorbs these observations that it has what those observations are it might be you might not pay attention to certain things which in one environment might be less important than another one example of a loss function that's um particularly natural and relevant in probability theory is known as the relative entropy or the coolback library or KL Divergence and this measures the difference from one well it kind of measures a distance or Divergence from one distribution to another on the same space and we'll see in a minute that these things that these relative entropies also themselves satisfy a nice compositionality property which again comes out of a gridity construction and so it its structure is also kind of captured by this abstract categorical machinery one of the reasons the relative entropy is so important in a proximate inference if you may know or have already guessed is that if we minimize the Divergence say between this approximate posterior which I've written as C_{Prime} π and the exact Bayesian posterior C_{dagger} π effectively what that does is it pushes the approximation closer to the kind of true or the optimal um the optimal inference and so often people start an approximate inference

process by trying to figure out a way to minimize this quantity but you can see here again that we have this dependence of the this quantity this Divergence on the prior because of course the inversions depend on the prior and on the observation so why is the thing that goes into yeah the system like this we have this diagram before and so the actual Divergence in any particular context depends on what the observation was and so if what we want to do is compose these um these loss functions for one say one little neural circuit and another little neural circuit we should figure out a way to pass the the information of this you know this this observation and this prior into those loss functions if we write down the type as I said at the beginning this kind of discipline about types is one of the things that makes category Theory to me so useful if we write down the type of this function we'll see that it corresponds to um one of these effects or co-states that I drew earlier as long as you think of it as also depending on this prior so what I think I call that a state dependent effect effectively it's a state dependent channel that lands in this kind of trivial kind of empty diagram object here what that means if we interpret this type in the mark of Kernel language that I introduced earlier is that it's a function that takes in a prior and then an observation and it gives us back a number and in particular it could give us back this relative entropy and one thing that is known about the relative entropy and in fact is quite easy to check is that it also satisfies a chain rule the chain rule has this form but I you know don't need to go into details about what it means what what it yeah except to say that effectively the composition of the relative entropy for one lens if we think of this part and then this part has two lenses this we might have a relative entropy which lands in the positive reals for here and a relative entropy up here and another one up here this chain rule says that the composite if you apply this kind of lens composition rule to it the composite relative entropy or the relative entropy of this composite lens is equal to the kind of lens sum of the relative entropies of the kind of independent lenses and so that's that's this locality that I was referring to so this is a nice equality again this is only a particular example of a loss function and because if you notice here sorry I'll go back and slide just momentarily you notice here that the relative entropy depends on this exact inversion well that means in order to be able to compute it you need to know what that exact inversion is and actually often the reason we want to do approximate inference is because Computing the exact inversions is quite hard you know we don't want to try to compute that thing and so we need to do something else and so these other loss functions are available um and the relative entry is one example but we could we could sort of pair a lens with another example of a loss function and I've given a composition rule for lenses and I've sketched a composition rule for the loss

functions and so these things form these kind of pairings of lenses with loss functions also form a category and I call this category the category of statistical games and the reason I think of it as the kind of game um is twofold the first is that it you know shares a lot of structure with categories of economic games as I've indicated already but it also I you know it also captures something that's quite nice you know to think about the use about these processes which is that you know the way we improve our inference is by playing this game of how can I do my inference better so that's this statistical game that you play that they say these neural circuits play what we've done here is we've attached loss functions to lenses and so just as we had a vibration by attaching the the inversions to the channels themselves we've done this attaching thing again so we've got another vibration and so we could ask okay so what are the sections of this vibration so what are the ways of assigning in a kind of nice compositional way to each lens a loss function um there are various examples so one of them as I've said is the relative entropy that's particularly nice because it makes this thing an equality but there are lots of other examples that people use in applications because Computing the relative entropy is difficult so we've got things which is like the free energy which is a abound on the relative entropy and we've got things which is we've got something which is slightly different we've got the maximum likelihood which is a way of saying okay well how do I improve my predictions if I don't care so much about the actual the inferences that come out of those the free energy I should know is basically the sum if that is defined as the sum of the the relative entropy and maximum likelihood and it's nice that if you make that sum not only do you get something something that's a bit easier to compute but you also find a loss function which kind of encourages you both to improve your predictions that's the maximum likelihood thing as well as improve your inferences in machine learning a process which does both the Improvement of predictions and of inferences is called an auto encoder if you've heard that word in Neuroscience often it's necessary to make further approximations to get to something that seems to be like or possible for the brain to compute um and so one such approximation is known as the LaPlace approximation and if you apply that to the free energy you get another loss model that is as I say lacks in the sense that there's some other term here to make this equation hold so these things all form loss models possibly lacks lost models and that means that they are nice kind of local compositional ways of attaching um loss functions to these Bayesian lenses and indeed um this laplacian free energy loss model is the kind of family of loss functions which underlies uh say the most prominent models of predictive coding so we've started with these Markov kernels or channels as I call them we've attached um

inversions to them first to give Bayesian lenses and then we've attached loss functions to those lenses to give statistical games but there's not much freedom in that process to actually improve the loss functions because once you've fixed a lens that's it the only thing you can change is the input either the kind of belief the prior belief you start with or the input you get from the world what if to improve the loss function you also need to change the lens itself so we can use the same kind of idea that gave us State dependent channels here and parameterize the the Bayesian lenses and the loss functions to give us parameterized lenses and statistical games and in machine learning in particular it's updating these parameters is what happens during the training of models and in models of neural circuits um it's we can think of the sort of synaptic weights on the neurons as the parameters and parameterizing a structure is something that is kind of um quite well well known in category Theory and particularly in categorical cybernetics um and the way I like to think of it is as the kind of choice of process so you get something which takes in a Θ and gives you a morphism and you can compose those morphisms themselves and then learning amounts to kind of improving that choice process you know this is just a particular it's a general morphism in the category $\mathcal{C}$ but we can make this category the category of statistical games and then we have a statistical game which depends on some parameter and then we can optimize the loss functions which are of course part of the morphisms of these games with respect to those parameters and we can do that you know with respect to each parameter on each component in this kind of local way as neural circuits seem to do and exactly this is how predictive coding models are obtained so as I said we've we've paired predictive Channels with State dependent um these exact inversions constitute a section and other other sections other lenses represent approximate inversions we can pair the lenses with loss functions in the same kind of way to give statistical and this formalizes the chain rule of the relative entropy the free energy is another example of one of these loss models and then we can make space for learning by introducing I'm aware I don't have much time left and so I won't spend an awful lot of time on Dynamics I just want to say that the same kind of idea of filling in boxes so earlier I drew a box and I added a parameter into we can use that to give us a notion of kind of compositional dynamical system and I think John's going to talk more about this later possibly I mean a lot of what John has done is about dynamical systems um and so what we can do is say okay well we've got a category that represents the interfaces of our dynamical systems and then we can use the morphisms of that category to represent how those systems are kind of wired together and this this this category of interfaces and wirings is something I'm going to call Int and then on each interface so this is

the interface which has inputs X and inputs y we have a category of dynamical systems which take in interfaces X and Y and so here we can fill in one system and here we can fill another system and you can see that this is the same again the same pattern that we saw earlier of an indexed category and so here the re-indexing is the rewiring of systems and so what we can do is apply this wiring to the systems themselves to get a sort of bigger system on this composite you know on this on this out for this outer box a kind of composite then we can Define dynamical systems which emit the data of Bayesian lenses I call these these systems cilia because they're like the ciliary muscle of an eye they control the lens um I won't go into details about the structure I just wanted to write it down for reference but a cilia the Cilia have the same type as the lenses um but they're now sort of dynamical lenses and so a psyllium of this type that goes from XA to YB it has a state space it has a forwards output Channel that's parameterized by the state space uh backwards inversion that's also parameterized by the state space and then an update function or an update map which is like a stochastic dynamical system which takes in a state space and an observation and a prior belief and gives a new element of the state space and so that's like the the sort of updated parameters that we used for learning and these things compose by composing the lenses together and by forming like the the product of the of the um of the parameter spaces so you have a parameter on each on each bit of the composite and this so this story it starts to give us a general recipe for predictive coding systems which we could think of as like local approximate inference systems you know we have parameterized kernels we form lenses then we make statistical games the first is the thing I call an inference system and the second is a lost model and then we've gone from a parameterized statistical game into this dynamical category they say doing optimization as David mentioned at the beginning on the loss functions and so a process that starts in this category of stochastic maps and ends in dynamical systems that do inversion is something I like to call an approximate inference Doctrine it tells you a way to do approximate inference for each channel in your original category I will end briefly I just want to say finally that this is all about approximate inference effectively perception we can extend this to active um by saying that the CIS the the agent say is trying to predict not only um what it perceives but also what it should do um effectively predict everything on its on its interface the inputs as well as the outputs in the literature in neuroscience and um related areas this interface is almost known as its markup blanket and so we can also package up systems that do this kind of prediction into a category or if we call the interface P we get a whole category of agent Agent P of Agents with that interface to do this kind of prediction and indeed we can rewire these

things according to wiring of the kind I mentioned earlier so you could think of that as like if you get in the car like the your your sort of interface changes because you you know you're you're now contained within the car but the car itself has some intelligent Behavior you could think of a corporation which again is made up of lots of Agents but it itself has this legal personhood and so at this point we start to consider how agents themselves compose um and this kind of gets to some subtleties which are kind of more philosophical questions um in particular that agents they need to predict not only how the environment causes their sense data but they also need to predict not only their senses itself but also how the environment causes that sense data and that's something a bit like theory of mind and this is something that's necessary to make this structure compositional um and then once we have this you might think okay well two different agents May disagree about how how they're wired together and that that we need we need a way of sort of coming to overcoming that that sort of disagreement um this is really a sign that these Bayesian models are kind of inherently subjective and there are various ways we can deal with that um in mathematics there's a lot of stuff a lot of work done in cohomology to measure disagreements of various kinds um and we can use diffusion to sort of smooth out these disagreements or we could just have a higher level agent which imposes a resolution to the disagreement and because of the compositionality of all of this we can do this or do each make each of these choices in a kind of nicely modular way as I say the choice depends on the situation at hand and so finally this this kind of points to the kind of broad ambition of what's known as the free energy principle which as I said at the beginning asserts that kind of all adaptive systems um at least that maintain their interface can be understood as performing approximate Bayesian inference and at least I don't believe that this claim although it's been stated and there's lots of evidence has been supplied to substantiated I don't think it's been sort of conclusively established and I think using categorical tools and this formalism of compositional Acts of inference is a nice framework in which it might be possible to do and so what I've talked about is a mapping that goes from Agents as say statistical models to dynamical and the free energy principle claims that there's a mapping in the other direction which goes from sort of dynamical systems in say of other sort of physical kind to um agents or statistical models in a world that kind of has a more sort of biological flavor and we can sort of think of this as a kind of mind-body dualism and in particular one way that we might be able to formalize this claim is as an adjunction which is something again I think David mentioned at the beginning thank you I know that was quite fast but I wanted to cover a lot of ground to say how how there's quite a lot of you know

great stuff to be you know found by using categorical methods for complex systems like the brain yeah philosophy talks they sort of always leave a very long time to for the audience to sort of Slash it the speakers arguments but um obviously we're not we're not in that domain here but perhaps if we have one one or two questions which which actually I did I did there was a thought because we've got somebody here who works on explainable AI PhD student and what do you do so so say you've trained you've trained a neural net up and you've got these sort of different different levels if each level is to be seen as a Bayesian lens of some kind I mean how do I get it what is what is the type that is the output of one layer and the input to the next layer how can I get to understand what it means it sounds like you've already got the type articulated doesn't it if you're if you're confusing these yes so that's I think that gets to a point that the category theory is kind of more about composing systems and decomposing them so I'm assuming yes that you started with um two lenses that represent say your models of small parts of the system and you're trying to use that to build up a bigger system but yes if you have a kind of big complicated neural network um there might be various ways of partitioning it into smaller components to try to understand it but I'm not sure um I have a kind of nice Universal story about how to do that right now um I'd love to have a good connection from this to um to that kind of explainable AI but I think a sort of less ambitious aim that we start to get from this kind of toolkit is one that says okay well at least we know how our systems are built and we know what the different parts are if we work like this if you look at machine learning code as it is it's often full of lots of you know very informal commentary because people need to sort of be careful about where certain bits are which how the information flows effectively which index goes to which other index um and so I think people actually don't understand the models as they are right now having some types on those interfaces might make that process a bit easier okay great so thanks thanks very much always one one question from we've got Mark here from a PhD student yeah thank you so much for that it was a lot very very intense and I really did appreciate that um there's uh the there's something that I want to get on first of all I like the uh an explanation of the um impregnateness of the whole project uh um but uh the the question I I want to ask you is uh it doesn't seem like there's going to necessarily be a clear uh demarcation on the biological side the ancient biology side to uh the uh the process of uh of um of uh going from uh um uh Bayesian lensing to the Cilia I can see institutions and uh I think you hinted on that corporations and uh uh uh um uh the you know the uh the focus of uh of uh at the Cilia level might end up being at uh uh so I want to know what your your thoughts on that are yes it's

very interesting so I yeah I this is a this is a kind of scale free theory in some sense um I don't mind I don't think of a particular neuron as being more like adaptive than a whole brain or a corporation they can all be understood as doing this process at least on this account um and so there's not nothing to say that say a corporation is somehow less like biological than sort of a single or multi-celled small organism and in fact I think what we see of the process of the evolution of life itself is a kind of growing um sort of growth of complexity like you start with these single celled organisms you get multicellular organisms you get you know you know simple plant life animal life and now we've got societies and in biology you have smaller societies like ant hills and those things are all kind of sent you know sensitive sensible conglomerates of their own kind and they are adaptive um and I'm trying to capture this process in its generality um yeah corporations are part of that but you know nation states and the whole world is one two um so I I'm not trying to make a distinction between biological systems as they're commonly understood and these kind of broader economic systems I think of those things as all kind of ecological in some sense and to be honest I think economics could do a lot better if it started to think a little bit more ecologically too thank you so much thanks very much TV so we've got a few minutes to set up uh what have we got three four minutes till we get to the hour and uh and we can hear from John hi hey John how's it going oh yeah great can you can we try to try out the uh it worked with Toby so it should work with you if I get the people and then I find you and then they can presenter so that should work so if you share now they should okay let's see so I told you I'm a complete novice you're a Zuma another another teams are so how do I share does he not have a top right that was um yeah Fair tray icon okay yes it says open share tray okay that's another roundabout way of saying you want to share okay okay we've got okay so we're having oh yeah okay yeah I think if we close that side then that's pretty good let's listen yeah so we're good good from our end is everyone online is okay well let's see so we've got um two weeks till we begin okay great how's it going I'm woken you up too early in California uh not too too early what is it it's nice yeah it's yeah uh that's the problem of living on the very west coast of the world is that everyone else wakes up too early but I like the miracle we can we can do this sort of thing right I was gonna say I really like the ability to give talks without leaving my house so definitely worth it so yeah you were wondering about recording we couldn't still be going okay I'm gonna do it again I know it is in the process of recordings that's fine okay great uh so we'll wait till did I quit it 10 minutes to the hour um yeah I mean you know it's funny so philosophy talks I remember the first time I gave a talk to um you kind of you know you're expected to do about two to one

ratio of talk to questions and that some places even worse than that and then you go there and the Massachusetts wouldn't have really good questions to put you 95 talk isn't it and five percent questions but okay well I'll try to I'll try to leave time for questions I should okay yeah I'm raising lots of questions still okay okay so I guess we're there so let's get started so uh well gosh what an influence John John had in my life uh there I was a fledgling philosopher trying to uh interested in in mathematics and and gone through his writings in the in the Long Way a long time ago in the 1990s gave me the confidence to think that you you could get access to the Forefront of of mathematical and phys scientific knowledge through his writings I mean what gosh what a what a joy that was to to engage with his writings back at that time huge influence on me and you know a bunch of people will know that we started a Blog together a while ago we had a lot of conversations over over a great many years so it's great pleasure to have him speak to us here today on on the play category Theory so thank you John well thanks very much thanks for inviting me here uh the influence was reciprocal uh it's really exciting to think that what one is doing in mathematics could actually be interesting to a philosopher so that uh one isn't just like fiddling around with obscure equations uh so I want to talk about applied category Theory which is the latest turn in this game uh and I'll talk more about the history of the subject and what people are trying to do then uh any mathematics per se I've given lots of talks about the mathematics and so of all the people who I'll be referring to here so it's certainly easy to click on any of the links in my slides and get into the actual mathematics but I'll try to not say as much about that as I sometimes do we often have this picture of different disciplines or subject areas talking to their neighbors and information flowing between pure math which is at some very abstract and uh to say the business World which is maybe at the very concrete end in stages with each discipline talking mainly to its neighbors so care mathematicians talk to applied mathematicians they give suggestions to applied mathematicians and they find out about problems from applied mathematicians and learn techniques from applied mathematicians and similarly for each link down the chain with each uh discipline talking to its neighbors and in this model which of course is very oversimplified and and you can easily find a lot of faults in it in this model we could think of category Theory as the purest of pure some super pure mathematics that only can talk to mathematicians who then relay their insights down the chain to the other more practical subjects I think that's what a lot of people implicitly feel about category Theory or at least used to uh and so when people started talking about applied category Theory people laughed because it sounds like an oxymoron it just goes completely against this picture here however I

think this picture is inaccurate for one thing a very important fact is that starting maybe around 1980 or so it became clear that computer science really cuts across this traditional flow of information computer science interacts almost directly with every single piece of this chain computer science has a big relation to category Theory which I'll explain but it also has a huge relation to other aspects of pure mathematics and Applied Mathematics and so on down to business so business businesses will need to keep abreast of developments in computer science and so on basically we now communicate and think with the help of these machines called computers that we have devised to help us and and they changed the whole game at every single step of the way now I have this crazy idea that maybe category Theory actually really should be thought of in a somewhat similar way as interacting with all the different links in the chain from the pure end to the applied end rather than being tucked away at the super pure end of the spectrum now at first it seems ridiculous and are laughable I mean for starters category theory is sort of nothing compared to computer science everybody's walking around carrying cell phones um nobody's even heard of categories by comparison so category Q is an ant compared to the elephant of computer science however I'm thinking about what could be uh not what yet is and secondly when you think about how computers have unified the whole chain it's done so by means of the internet which is an enormous Network and category Theory isn't nothing if not the mathematics of of networks or the science of networks it's other things too I shouldn't have said it quite that way I mean it's it is the science of networks but it does many other things too uh so really devise good methods of getting computers to do what we want you need a way to think about them and I think category theory is one of the important ways to think about them so you may think this is ridiculous still because for example if category theory was talking directly to business you'd expect that people would be going into business during category Theory or something like that setting up category Theory businesses that sounds sort of silly but in fact it's true so in fact people are setting up category Theory businesses and that's part of the kind of thing I want to talk about I want to talk about how category theory is now being used at all uh steps along this spectrum and for that reason my talk will be less focused on the math than on the history of how category Theory became applied giving you a flavor of how it's being applied and pointing out some of the challenges that show up um uh before I can dive in I should say that this history is going to be extremely self-centered I'm not really in a position to write a good or talk about a good history of category Theory so hopefully someone who eventually tries to write a history of Applied category Theory will listen to this talk and fold it in as one data point uh I'll talk about a lot of

people but I will also leave out a lot of people and I really apologize to all the people whose work on applied category Theory I'm not mentioning uh there's just no way for me to do a fair treatment of this big subject okay so I'll start arbitrarily in 1963 when Bill of beer introduced a concept called functorial semantics so it's a very general concept where you have a category that describes uh syntax describes ways of writing things down and then you describe the process of interpreting those expressions or giving a meaning to those expressions the functor so from the functor will send expressions to their meanings but a crucial aspect of this whole setup is that there's not one god-given functor F that does this you can choose various functors from the category C to various other categories he uh because there are different ways to assign meanings to expressions and that flexibility is very important and it became important in computer so for example there's a nice there's a way to think about it we're a sufficiently well behaved uh programming language may give a category where you think of the objects as types or data types we've already heard about that idea uh so types x y z Etc and then morphism between objects like morphism f from X to Y we think of it as a program that accepts data of Type X as input and outputs data of type y and then the fact that you can compose tells you that you can hook up two programs to get a bigger program where the output of the first program becomes the input of the second and like that that combination is a program in itself so I could say this like the relationship between the Lambda calculus which is sort of an idealized programming language simplified programming language and Cartesian closed categories uh understood by labik is an example of this Paradigm of using functorial semantics for uh for computer science but here I'm only so far talked about the category C I haven't talked about the uh the functor so what the functor could do well we could for example take a function from our category to the category of sets not each data type X to a set and that would be the set of possible values of the of that data so like you could have a data type like real numbers and then definitely map it to the actual set of real numbers uh and then each program gets mapped ideally to a function and the key factor of a functor is that it preserves composition so that composing programs would compose the would give rise to composing the functions they compute and the identity program which does nothing at all you interpreted the Sumter f it gives you the identity function which does nothing at all on a set so that's a that's a paradigm that became important in theoretical computer science I wouldn't say that most uh computer scientists know or care about functorial semantics but in a certain community of the more mathematical theoretical computer scientists this idea has become quite important and there are certain languages like Haskell for example and others that that

really take advantage of this uh Viewpoint in particular Haskell uses a category theoretic construct called monads which has sent many uh generations of young computer scientists running for help from a category theorists to try to figure out what a monad is I don't want to explain what a monad is it's a category theoretic construct which like many other constructs uh become important when you start thinking about programming in this punctual synaptics way but in the 1980s something very different happened uh particle physicists realized that any Quantum field Theory specifies a category and now the objects are roughly speaking collections of particles and the morphisms are roughly speaking ways for particles to interact and turn into other particles and in fact Feynman Drew pictures called Feynman diagrams for describing how particles interact with each other this was a bold move because although it seems very intuitive here you have a proton and an anti-proton coming in from the left and then they Collide and all sorts of other particles shoot out uh in reality particles are not little dots moving along little straight lines quantum mechanics says that particles are wave like in character uh so when Feynman introduced this pictorial language you've got a lot of opposition from older uh Quantum physicists like heisenbergen and Bohr uh and yet it turned out Feynman was correct in doing this what finding diagrams actually are later became clear much later is there pictures of morphisms in specific categories and each individual Quantum field Theory gives a category and then that determines what allowed Feynman diagrams you can write so if you have a theory that doesn't have quarks in it you're not allowed to write this q line for for quark and then what do you do with these pictures Well you your goal in in particle physics is to compute a bunch of numbers like what's the probability that if I smack two particles into each other that some other particle shoots out so for calculating numbers we need uh to interpret our finding diagram that is in the functorial semantics attitude we need to functor from our category to some other category which assigns meaning to the finding diagrams in this category we could take to be the category of Hilbert spaces a widely loved mathematical structure in quantum mechanics and so then it will this function will turn any Feynman diagram into amorphism in the category of Hilbert's faces and that's called a linear operator between Hilbert spaces and so you can look at what physicists are doing with Feynman diagrams uh as basically using them as a syntax which they then apply the semantics to to then grind out numbers now Feynman probably would have laughed at this whole business because he knew what he was doing he didn't need any category theorists to tell him what what he was doing but by the 1980s particle physics had become elaborate enough and deeply connected with enough pure mathematics that this attitude of thinking of upon field

theory in terms of functorial semantics became a really profitable attitude you could do things with category theory that particle physicists were struggling to do without category Theory and so it helped and this led to a whole line of work uh trying to relate category Theory to quantum mechanics and in 2004 Samson Abramsky and Bob Kirby wrote a very influential paper showing that a certain Quantum process that goes by the name quantum teleportation and also other processes could be understood using category Theory and using diagrams that resemble Feynman diagrams but here the edges in these diagrams which now I guess we're reading from top to bottom rather than across the edges are not describing Elementary particles but just states of an arbitrary Quantum system so it could be an electron or it could be some larger system and I don't want to try to explain this in detail but there are but these are pictures of morphisms in some and they're equal to each other and these morphisms describe some process where one participant always is called Alice in cryptography uh communicates a message to another participant it's called Bob and in quantum teleportation you have the superficial appearance that you're communicating uh faster than but in fact what's really going on is that you've laid the ground for the communication by some classical process that has already transferred some information and then the rest of the information uh is transferred on the basis of that in any event quantum teleportation was a sort of magical wonderful thing that you can do in quantum mechanics and this gave a kind of category Theory theoretic account for how it's working and that was quite exciting to people because it took the quantum applications of category Theory out of the realm of elementary particle physics and into a new realm that was developing which is quantum information Theory or Quantum computation which has the potential for being a quite practical thing and indeed already a Quantum cryptography is a is it is used so on the basis of this and other work and also on the basis I should say over a huge amount of Charisma Bob Kirby and Samson Abramsky built a huge group of people at Oxford sometimes called the quantum group working on category Theory the foundations of quantum physics and Quantum computation one interesting thing about this enormous group here is that it's in the computer science department so not a mathematics Department not even a physics department but a computer science department here's Samson Abramsky here and here's Bob Kirby looking sort of disreputable over on the edge that's completely intentional he's like that and here are many many people including I see Jules Hedges who has mentioned uh in the previous talk that you were example categorical cybernetics um so Circa and co-authors developed this diagrammatic method for explaining quantum physics and they eventually decided it would be very good to write a textbook on this

and so here's one of the two textbooks that Bob Kruka helped write on pictorial methods of understanding quantum physics and the one interesting thing about this book so first of all I recommend this book if you're interested in learning category Theory from a unorthodox and sort of easier approach than the conventional one because it's very pictorial uh but I mentioned it because part of what what you wound up doing is hiding the category theory in the sense that you don't run around saying category and functor and all sorts of other jargon like that a lot instead you draw pictures of processes and you explain these pictures in a Hands-On way and you explain how you can stick one picture onto the other which is composing morphisms but you don't say the word composing morphisms necessarily because you don't need to uh and this is one of a generation I should say a new generation of category Theory textbooks that are part of the applied category Theory movement so if a category theory is going to reach directly to many different communities at different stages of the knowledge chain that I drew at the top of the talk uh it means that you can't force people to first become mathematicians before they then start doing things with category Theory the traditional approach to category Theory treats category theories like some finishing school that you engage in after you've already learned abstract algebra and this and that and algebraic topology and so on that's just too onerous a path to category theory if you're trying to get a computer scientists much less business people to get some inkling of category Theory and it's not really necessary to take that elaborate route because category 30 is something so fundamental that it can be explained more directly I especially recommend Eugenia Chang's book *The Joy of abstraction* which is a really wonderful introduction to category theory for people who have not had much prior exposure to mathematics at all and it goes all the way up to the Oneida embedding theorem thank you so in 2021 Perka actually left Oxford University and he became the chief scientist of a company called Continuum which uses category theory for Quantum Computing and also natural language processing this is a picture of him on the Continuum web page as you can see he's trying to quite hard not to fall into the stereotypical academic image here he's a great guy by the way you shouldn't mind me teasing him like this uh I think though that it is important it's having people with personalities who are willing to break out of the stereotypical mold of math professor is important in this whole category Theory application movement uh I should say that there's a lot of interaction between Quantum Computing ideas and natural language processing ideas which I won't really have time to get into uh to some extent they were visible in the last talk but if you want a real emphasis on the natural language processing aspect *The Works of* uh Tai Dena Bradley are very good she's another newer explainer of category

Theory it's a non-category theorists and she is also working uh in a company that uses Quantum methods and categorical methods for natural language processing but Ontario semantics is extremely General so we it's not just applicable to the topics that I've already mentioned so in lots of areas of Science and Engineering people use a syntax where you you draw diagrams and then a semantics where you convert those diagrams into math which is very often the systems of differential equations you're describing how things change in time you're describing dynamical systems uh so you use differential equations but the interesting part is that hooking up diagrams to each other is actually composing morphisms in a category and the fact that you're you're working with frontorial semantics says that when you compose those morphisms in the syntax category your functor will map it the resulting composite to the composite of the two systems of differential equations that each picture separately is mapped to so in other words there's another category lurking around here where the morphisms are actually systems of differential systems of differential equations with some specified inputs and outputs to surface as the source and target of your morphism here are some different pictures you've seen a bunch of these types of pictures I hope so electrical circuit diagrams are are one of the earliest and very widely used this thing here is a called a petri net and it's a petri net of a particular model for the AIDS virus and its interaction with blood cells infection of of white blood cells this here is a very small uh diagram in the subject of control theory where you have some process in an industrial plant that you're trying to keep under control and so you take feedback you read off information from that process and you use it to feedback affect the the process itself to keep it on track I mean the very simplest example here would be like a thermostat but in in industrial chemistry they become much more elaborate and so there's a whole diagrammatic language of control theory that you can read about in textbooks on that subject and this is a little diagram of some molecules in molecular biology molecular biology or biochemistry they have so many different diagram languages that in fact they've set up a committee on to standardize the diagram languages so that people reading different biochemistry papers can look at the diagrams and not have to be explained each time what the diagrams mean but it's sufficiently Baroque that they uh that the committee wound up choosing three different diagram languages because they they need multiple languages to describe different things they're interested in so these are very widespread and the people in these different subjects don't think of what they're doing here as category Theory they don't think of it as mathematics they think of it as this is the stuff we draw to understand the differential equations that you can convert these pictures into the equations of the math but

these pictures are our own private little trick for understanding what's going on but I claim that these pictures are actually also mathematics and they're these diagrams are morphisms in various categories which are using punctual synaptics to convert to differential excited that had to be true and I wanted to work out the details around 2012 so I asked Brendan Fawn who eventually became my grad student to tackle what I thought would be a very good first example namely electrical circuits because everybody knows about electrical circuits and the mathematics of them is in substance very well understood linear electrical circuits for example made out of just resistors or maybe capacitors and inductors as well I asked them to create and study a category having open electrical circuits as morphism so these are electrical circuits which have certain things hanging out which serves as inputs and outputs I think in the last part they are called cilia so these are just these dots here um so it took forever to do it so it seemed but it turned out that what he was doing was not just answering the question that I asked but in fact uh answering a much more general question he invented a general theory which he called decorated Coast bands for handling this problem but also all the other problems that I described that is essentially all the other kinds of diagram languages all right showed can be understood using the theory of decorated co-spans although the details have to be individually worked out in each case this decorated co-span formalism is a is a recipe for getting categories of this General sort and so naturally it did take him a while to do a good job of it and he finished his thesis on it in 2016. I won't attempt to explain decorated post bands any more than I just have I've given lots of talks on decorating post bands and you can see on YouTube me giving talks about them if you're interested um I just want to mention one little thing is that well first of all here's Brennan Fong and here's another student of mine Blake Pollard who together with Brendan applied this theory of decorated close bands to the second example that we tackled which was Markov processes honestly so so Blake did a lot of work on Markov processes uh and another thing I want to mention is that uh Brendan went to get his PhD at Oxford so he was in the enormous group that I mentioned and I believe Kirko is his de facto it was his day jury advisor but in fact I was the the de facto advisor telling you important to Brendan about this this problem uh and so nonetheless it's really important that this large body of people that uh at uh at Oxford is is spreading across the world doing many things and Brendan is one of them so after doing this doing his PhD uh Brendan went to MIT in 2016 to work as a postdoc with David Spivak so spivack is another very important figure in applied category Theory so among other things he had helped to create a language for databases called functorial query which has now become the basis of a company called Connexus AI with Ryan

wisniewsky as a as a head of that company but David Spivak also involved approach the databases really relies heavily on functorial semantics so the idea is the database schema is is an object in some category and then by a functor you can map it to a sort of populated or inhabited database schema where where the the your table of is actually filled with entries you can imagine like taking a spreadsheet and actually typing in the entries that's one possible uh interpretation of of the abstract database that has not yet been interpreted with with specific people's names and numbers and in 2014 David wrote a textbook on this idea and others called category theory for the Sciences so I also highly recommend this as yet another uh of this new generation of category Theory textbooks it's trying to reach out beyond the traditional confines of the category Theory community of Highly pure mathematicians toward other audiences and in 2010 2018 uh Brendan Fong and David Spivak wrote another textbook explicitly on applied category Theory which is uh fondly called seven sketches but its actual title is here earns larger uh emphasizing the principle of compositionality in many guises and they taught a course on it which you can see on YouTube if you click on this link in my slides and the textbook is also available for free on the archive as well as for purchase and I'm mentioning this stuff about things being available online for free partially because when you're trying to reach out of the pure math realm into the rest of the world and teach the rest of the world category Theory you quickly realize that making things easily available for free is going to speed up the process a lot and so I think there's a general commitment so far on the part of a lot of Applied category theorists to make things open source and uh easy to get at now in 2018 my graduate student Kenny corser there and I came up with structured Coast Fans which is a simpler alternative to decorated Coast bands turns out to be not quite as flexible as decorated Coast Fans but in many cases either formalism works and in those cases I'd say structured co-spans are simpler in 2020 Christina vasilicopolu who's standing to the left there uh helped us understand how decorated and structured Coast bands are related for a while I thought structured Coast Fans would replace decorated Coast Fans but it turns out we we now see that they can't cover quite as much territory and that we really need decorated co-spans in some case cases so the full story is somewhat elaborate right now uh I've given talks about that but it's it's actually very pretty so by now structured and decorated Coast Fans have been used to study open systems of many kinds so by open systems I mean systems that have loose ends that are meeting systems that are morphisms in a category that you can stick together or compose to get bigger systems and there were open graphs are just to Simply a graph with Loose Ends it's very combinatorial structure that's a very simple example open Petri nuts is a more useful example here's a picture

of an open Petri net where you have things describing are what they're called called places that could represent like populations of different kinds of chemicals or organisms and then you have these blue boxes called transitions which describe how things can come out of various places and turn into things of other kinds um if you attach numbers to these transitions uh describing the rate at which the transition tends to occur you get something called a open Petri net with rates or maybe an open chemical reaction network with rates and my student Blake Pollard worked on those open electrical circuits I've mentioned open Markov processes I mentioned open dynamical systems those at the systems of differential equations interpreted as morphisms in some so there are papers on all of these and if you click on these Brown things you'll get to those papers so at a certain point I thought that maybe someday all of this would lead to some unified language of the Sciences and my ambition which seems very limited in retrospect was that by interacting with Scientists maybe eventually this unified language would help scientists in one discipline talk to scientists in another discipline and share tricks of various sorts but then something quite different happened which is much more exciting in my mind namely in 2019 James Fairbanks is an engineer and Evan Patterson who was a statistics PhD began developing a computer system called algebraic Julia for it's a system for high performance scientific computing built on the language called Julia which is designed for crunching numbers really fast algebraic Julia lets you program it using constructs from category Theory so in algebraic Julia you can talk about categories functors and all these other things that category theorists like to talk about and write your programs using those contents so it's a very very high level system but coupled to the raw computational power of algebraic Julia so now we're attractive revolutionize scientific Computing in this way and in 2020 Evan implemented structured co-spans in algebraic Julia and then using this system uh for the example of open Tech units he and Mika halter reconstructed some of a covid-19 model which was currently being used by the UK government so the UK government had a very elaborate model of covid-19 with many populations of different people with different conditions possibly living in different locations Etc uh and then I can start the populations of these orange boxes and able to turn into other kinds of people via these transitions in blue whole model is much larger than this this is just a tiny fraction of it and what they wanted to demonstrate was that you could build the whole model by building smaller models and pieces and then subsequently sticking together those pieces which is in fact something that had not been done before models were sort of had to be constructed in whole cloth by essentially by one person keeping track of the whole thing but this is a new approach compositional epidemiology and

so it attracted the attention of some epidemiologists especially since covid-19 was a very live issue at the time then in a different direction in 2021 Brendan Fong and David Spivak who had been remembered over at MIT decided to quit traditional Academia and start their own Institute called The topos Institute in downtown Berkeley it's an Institute for Applied Mathematics but with a heavy slant on applied category Theory and they hired a number of people including Evan Patterson and also Sophie Lipkin who's here who will appear later in the story and it's a real real incubator for our ideas on applied category Theory now and so this is another one of the sociological moves that I had to be made I think to bring applied category Theory forward I recommended Brendan not try to start an Institute right after his first postdoc but luckily he was much braver than I would ever be and he went ahead and did it and it's quite successful now so in 2022 Sophie Lipkin Mika halter Evan Patterson James Fairbanks and another person I haven't mentioned Andrew boss used to systematically build models of epidemia epidemiology using open Petri Nets with raids so here are some open Petri Nets the rates that is the number data is not being shown here and here they're being stuck together to form a larger open Petri net the way that they're being stuck together you'll notice is not end to end this is not the normal end-to-end composition that you see in a category this is something a little bit more General for sticking them together using a trick called an operadic composition is one of the things that they Incorporated in their algebraic Jewelers software which is this was done this previous work was done in a slightly hot basis but but this really used operads in the systematic way and then in 2022 also another team of people including myself used to algebraic Julia to create software for building models of epidemiology using another style of diagrams called open Stock flow diagrams well stock flow diagrams but we're making them open so here are some of those kind of diagrams being stuck together in an operadic style that is not end to end but sort of gluing them all together to form a bigger model so part of the point here is that when we started talking to actual epidemiologists we applied category theorists learned more about how they actually want to do their models and they didn't really want to do their models using pepinance primarily they wanted to use a different technique that they considered congenial called stock flow diagrams um so in fact there's a community of epidemiologists who use stock flow diagrams to model the spread of disease and it includes my co-authors Nathaniel Osgood and his grad student xiaoyen Lee and one crucial fact is that they did covid modeling for the government of Canada so they were requested by the government of Canada to set up covid models for the a number of different provinces and in the process they gained a lot of experience uh with stock flow models they

already had a lot of experience and they wouldn't have been asked to do it but they gained a lot more experience and they learned some of the problems of the existing technology this is a beautiful thing about stock flow diagrams which is that with a little bit of just a little bit of training you can read them and you can understand what they're talking about so a fellow named Hoffman has pushed the idea of community-based modeling where you go to a community that has some problem like a disease or some other type of problem and you help model it and you model it by drawing stock flow diagrams which here are hanging on the wall in a room in rural India and that people can look at those diagrams and comment on them and help you improve the model because these people the local people they will have the knowledge of what's actually going on that the you the modeler who just flew in from from Oxford or wherever may not have and here we see a woman here looking at the picture on the wall and this is actually from a story where this woman noticed that some ingredient was being left out of the model uh that these Village Elders hadn't known about or had thought about and so she helped fix the model so that's the example of what you can do and you'll notice that none of the people in this story here had to know category Theory but they were in fact uh drawing more systems in the category on the notes to themselves now most stock flow modeling in the commercial level or the hype or the government level is done using a piece of software called any logic which is very powerful but it has several big problems it doesn't let you compose models it lets you make a model but it doesn't let you take a bunch of models and stick them together it doesn't separate syntax from semantics it only has one semantics namely the differential equation semantics which is the most important semantics but there are others that you would like it has no support for what's called stratifying models that term was very confusing to me it's sort of charging in on this community what it means is taking a model and splitting One stock one type of group of people into several stocks so for example you might have a simple model and then you may say oh we should subdivide people according to age groups because older people who get the disease more more than younger people or something like that and you would like your software to make it easy to to do that um it had no support for collaboratively building models it's basically one person as the software and they make the model they could talk to other people but they're the one who's building the model and it's not free it's commercial piece of software and it's not open source so people can't get in there and modify it and improve it so our new work and to fix all these problems and I could say we're well on the way to fixing all these problems so the reason it's so important to produce models is that the models that get actually used in the real world are

immensely complex so here's a picture of Osgood and Lee's covid model this is a page in any logic the software this is the whole model as you can see it's an enormous web of of stocks and flows and such and you you really would prefer actually not to have to deal with this whole entity at at once you prefer to have it be thought of as being made up out of smaller pieces because mentally it is made up out of smaller pieces uh but but that's not captured by the the software unfortunately any logic just doesn't do that for you so we've not created software that does that it supports compositional modeling with the stock flow diagrams so it's called stock flow and if you click on this link you'll go to GitHub which is where computer programmers keep their open source software so you can get download the software and so it lets you create and compose open Stock flow diagrams and lets you stratify them that is subdivide stocks into smaller stocks for those of you who like category Theory I'll mention that that's done using a technique called pullbacks in category Theory it lets you choose among several different semantics functors so one of which is the differential equation semantics uh and of course it lets you solve the differential equations once you get them and then Nate Osgood and another student of his Eric redika went further and made a graphical user interface for stock flow so stock flow is written in this programming language called algebraic Julia to interact with it you needed to know algebraic Julia but this graphical inner graphical user interface called Model collab it runs on a browser and you manipulate the the model by actually dragging and dropping little boxes and drawing little edges between boxes still uh and because it runs on multiple browsers teams of different people located in different locations can a lot can collaborate to build stock flow diagrams so I could build my piece you could build your piece we could both save them they'd be saved on the on the cloud and I see there's this little thing called Cloud here uh that that's not what it is actually for them sorry that's a mistake uh but but you can you can say it's gonna laugh at that um actually clouds are these little funny things here uh but you can say you can save your models on the uh on the internet and so that other people in different locations can access them and then glue them together to form larger models so in the goal to really apply mathematics there are many levels because the next step is to actually train epidemiologists to use model collab so we have some things going for us so first of all eight Osgood runs regular training sessions which goes by the Quasi militaristic terminology favored in computer science of boot camps because they're supposed to be intense so it runs boot camps on epidemiological modeling so he is beginning to teach people model cloud second no understanding of category Theory or the programming language Julia is required to use model collab so it's really crucial when you're applying category three that you

can ultimately Black Box the category theory in it so that people do not explicitly need to know it to use what you're creating although they may be using it in some kind of implicit way uh and third it's free while any logic you have to pay for which should ultimately be an advantage but I should still say it will take a lot of work because when you have a generation of modelers trying to use any logic this new paradigm is not so easy to shape so as I just already said applying mathematics takes many steps and as I've moved from being more of a pure mathematician towards a bit more at least some some days of the week of an applied mathematician I realize how easy peer mathematicians have it you just write a paper you explain your idea you throw it out there and you hope that somebody reads it but when you're trying to actually carry out the full application of an idea you have to interact with many people with many skills persuade many different communities that what you're doing is actually worth doing and it's much more elaborate but I should also say very satisfying endeavor laughs so I'll conclude by saying that applied category theory is interesting in in many ways some of which I hope will be interesting to philosophers uh so some of these ways are separate from sort of the core mathematical ideas so one thing as I hope I've illustrated here by my somewhat self-centered history is that it requires new forms of communication and collaboration because you're talking to new types of people and luckily uh these are starting to come into existence so there's an annual applied category Theory conference that's first started in 2018 uh run that time by Brendan Fong and some of his other young friends another is a journal so there's a journal called compositionality which is again started by Brendan falling in some of his collaborators and friends this is an open access free to publish free to read Journal of Applied category Theory which I recommend to your attention and another is called the category Theory Community server which is sort of like a a chat site or a discussion site for Applied category Theory actually for all category three but it has a lot of Applied category theorists there uh this was initiated by my grad student Christian Williams and so these are different ways for category theorists to start talking to other kinds of people people who want to might want to apply category Theory or learn category three and I hope that these catalyze the the process done trying to outline here another thing is that category Theory raises ethical issues in ways that category theorists aren't used to pure mathematicians tend to feel themselves somewhat insulated from a lot of ethical issues because what although their math make it used for all sorts of purposes they don't consider themselves particularly responsible for how their mathematics gets used because it's so abstracted they sort of can blame someone else for understanding it and doing something with it but when you start doing explicitly

applied to category theory that sort of veil of uh protection is is ripped open and you see that what you're doing may have a fairly immediate effect so for example the biggest funder of research on applied category theory is currently the US military right now and also a lot of Applied category theory is being used for artificial intelligence or machine learning or large language models or so on so I'm not saying that those are necessarily bad things I'm saying that those are things that definitely raise serious ethical issues and so I'm worried about the danger that applied category Theory will wind up merely intensifying deal already existing patterns in society that is it will only make the richest and most powerful people even richer and more powerful I would prefer it if applied category Theory could do something really different really new this is one reason why I've tried to focus on epidemiology which is a little bit different than than some of the other uh applications of course even at the demiology ultimately feeds into the healthcare system so there are questions about that as well I would really like to use applied category theory for environmental issues uh but I haven't quite gotten there yet anyway sorry I just wanted to emphasize that uh you know ethics is the provenance of philosophers and this is a really interesting intersection of mathematics and ethics which I hope that people think about and another thing applied category three does is Revitalize what is a very old question which mathematical structures are best for describing and designing systems I emphasize designing because with the rise of computers whatever you can think about clearly enough you can make so it's not merely describing the external world it's you're also making worlds based on your descriptions and so this immense Freedom that you have in computer science that the question is what's what should guide you and it opens progress it opens possibilities for some really radical progress because these are some really brand new ideas that we're talking about here that make let us organize systems in new ways so I think there's a lot to think about here and I hope I hope some people here uh give it a try okay I think I'll declare it done brilliant you can see why I got drawn in by him 30 years ago it's uh his ways of conveying ideas so clearly okay do let's have some some questions then let me bring up the people perhaps if people online have some questions that they can raise a hand and I hopefully should be able to see or anyone here uh okay uh Mark's but I I do have a question and I'm I'm finding stuff that I've thanks for that by the way um this is really exciting stuff for me but I'm wondering if I'm biased in reading too much into this that uh uh that just works in my self-interest um I do philosophy of time and um I'm one of the uh the few people who defend the notion that time flows and uh um I see in your work uh but I'm uh maybe it's through these biased eyes that uh this uh this project is bringing back the idea of uh

of thinking of uh differential equations for instance as things that flow uh that functions or something that flow and I noticed the importance of diagrams as uh and these diagrams of uh uh uh are reminiscent of what organizations would call flow charts and uh um I'm I'm wondering whether I'm reading too much into this as something that might be an answer to the the static view of uh of uh in in in maths that came out of uh set theory and uh uh um thinking of uh um functions as uh as something that uh is static and not doesn't float so I'm I'm wondering whether or not I'm reading too much into it so maybe maybe that's the question sure um yeah the question of static versus Dynamic is as you know very well or extremely tricky question because I could take any dynamical system and when it's when it's done or when we imagine it as being laid out in time then we can think of it as a static thing with time is just another space like Dimension and and I think there's three it's very hard to like definitively break out of that uh so but nonetheless category theory is a generalization of set theory in a sense in which instead of just having elements in a set you have objects and also morphisms and the morphisms have are meant to convey some kind of dynamic that aspect of the system that is a amorphism goes from an object to an object and that's a very primitive sense of Dynamics there of the the start and the end and and and it brings a kind of dynamical quality to to the foundations of mathematics which is is a real thing it's not it's not just an illusion uh I mean even though you could if you ask a traditional mathematician what is a category and there's a oh it's a set of objects and a set of morphisms and some functions doing this and that right so you could say oh or just back in the same old Paradigm and that's what I meant by it's hard to definitively break out of that Paradigm I I think it's more in some ways more of a question of attitude rather than than anything else and anyway I think the interesting thing about the differential equations here may be the interesting new thing maybe the study of open differential equations which are differential equations where some quantities are explicitly labeled as affected by the outside world affected by the part of the world which is not modeled by the differential equation so uh traditional differential equations well no physics often like the study what are called autonomous systems of differential equations where you know everything there is to know once you know the differential equations uh where as open differential equations which I should say are also studied by persistent especially Engineers have certain quantities which are not determined by the differential equation they're regarded as being affected by the outside world and and that that on the one hand allows you to treat systems differential equations as morphisms that could stick together by sticking some of these quantities on the edge to some other quantities on the edge but it

also means that you're not just dealing with a deterministic system in the classic sense anymore because you have quantities that are affected by the unknown so so they it definitely changes your attitude towards differential thanks uh a question from Toby it's not so much a question sorry although I would love to I mean there's also there's questions I could ask but I I need to head off because I actually have to go give this talk again just now thank you very much for inviting me and uh I hope the discussion is profitable okay thanks baby uh I had a um I was telling the interests a lot of people around here is is the difference between when you get variables that are just sort of associated with each other there's some kind of statistical Association versus when you think there's some kind of causal connection between them uh and you might think you might think you're a much better grip on the world if it isn't just Association data but there's a you've got a you know the a causes B other ways of getting at this through your um or is it is it that you you I guess the modelers are kind of coming already with their sense of what causes wild and is there any way of representing the difference yeah so this particular line of work that I was outlining as you say I was mainly most of my talk was about model building where you you have your opinion about what causes what and how affect what how X affects Y and you build the model into that so there's a flip side of the whole um modeling process which I didn't talk about at all which is incredibly important in epidemiology which is you know inference so you you measure data and you try to establish causality and also various numerical quantities which you would then put into your model um so so there's a lot to applied category theory has to offer to that and I'm afraid I'm not quite as uh knowledgeable about that I will say that um Brendan Fong for his master's thesis uh worked on a categorical approach to uh Bayesian networks and so and and and that is subsequently grown into things like what uh Toby was discussing uh and so that should that's that's also part of a big story and another part of it is that there's a technique called uh particles filtering where you which is widely used by epidemiologists where once you have one of your models of the sort that I've been discussing you run it and you run it a bunch of times and you see what kind of outcomes it predicts but then you use that to uh to compare to reality and you use that to estimate the parameters that you'd be in your model so so I guess what you could say is that like what I've been describing was just sort of one half of a feedback loop of interaction between modeling and then comparing model with reality and and yeah so the whole thing should be should be carefully understood and I'm afraid I have sort of focused on one half of it sure yeah I've seen various diagrammatic techniques from category theorists and really get it kind of causal modeling I mean that exists out there doesn't it yeah it does and

that's important okay um well at this I think there's no further questions when we reached six o'clock anyway so let's just oh sorry was there one there was a one sorry there's dominance gone here you go yes sorry I'm casting around I have lots of things to say sort of trying to formulate them into the questions typical hi John uh um so one thing that in um I know in computer science is quite a big deal of sort of algorithms and data structures um and kind of getting the right data structure for the right task to get good performance I think you know there's a lot of things that category theory has to say about this um uh kind of describing data structures in in a more abstract way sort of polynomial functions and things like what I haven't quite sort of understood is is how much of that kind of work is going on in in the applied category sort of movement recently sort of linking with back back with computer science the sort of like programming aspects around what I would call kind of classical compute science of algorithms and data structures um do you know if there's much of much of that going on within the sort of act World um or is that um largely more back with the computer scientists um so the algebraic Julia Community is really trying to blend computer programming with category theory in a really rich way so that you can write software and you can say you know Define functor define pre-shift category blood so that that and that's just one of several different initiatives to try to bring category Theory into programming so not just have it help programming it but I would actually be part of the programming so that that's one thing there are lots of other things and I probably don't know the other things so I will just mention again this company uh connexus.ai uses uh categorical approach to databases so it turns out that the updating databases is very nicely described using categorical ideas like on extensions and um Mike Johnson in Australia has been serving as a consultant for uh for database companies using category Theory ideas uh for many years but now this company based on David spivex work is is also doing that uh and there are and there are a lot of other things I should say that there's just a lot more on the computer science side that I'm not very familiar with okay all right okay thank you very much okay so well here we've reached home so let's what we have to do is thank John very much for his for his very interesting talk [Applause] so I've recorded the session and we can you said you're thinking putting it online we could yeah I'd like to if that's acceptable trip yeah yeah let's do that good all right okay well thanks thanks ever so much John okay thanks very much bye bye everyone online bye I'll see you see you look like all right